Classifying Malware Families Using Graphlet-Based Topological Signatures from MalNet-Tiny

Hernan Barajas Tom O'Connor
University of Notre Dame

***Abstract*—This project explores whether malware families can be identified using only the structural topology of their call graphs. We extract 2–4 node graphlet features using ORCA, compute Graphlet Degree Vectors (GDVs), and construct Graphlet Correlation Matrices (GCMs) to form graph-level fingerprints suitable for learning. We evaluate both a coarse baseline based on simple graph statistics and a graphlet-based benchmark that captures higher-order topology, comparing classifier performance and feature effectiveness across families. Our goal is to determine whether structural patterns alone encode family-specific behavioral signatures.**

I. Research Question

RQ: Can malware families be classified using only graphlet-based structural features extracted from their call graphs? This study focuses exclusively on topology-driven classification without dynamic analysis.

II. Data Understanding

To ensure meaningful modeling, we first characterize the MalNet-Tiny data. For each graph, we compute node and edge counts, density, degree variance, and component structure. There are roughly 7 million nodes and 14 million edges across all graphs in the dataset, and an average of about 1,410 nodes and 2,860 edges per graph. The graphs are rather sparse with a median density of 0.002 and a maximum density of 0.20 in the dataset. We also examine the number of objects per malware family and note potential imbalance that may affect classifier behavior. To confirm that the call graphs are not random, we compare a small sample to Erdős–Rényi graphs with the same (n, m) and evaluate differences in clustering, degree spread, and small subgraph presence. Five representative graphs are visually inspected to understand recurring structures such as hubs, branching patterns, and modular regions.

III. Feature Extraction Pipeline

A. GDVs via ORCA

We use ORCA to compute Graphlet Degree Vectors for all 2-, 3-, and 4-node graphlets. Each graph yields a matrix where each node is represented by its orbit participation levels, capturing how the node appears within different local configurations.

B. Orbit Reduction

Since some orbit features contribute little information, we remove those with near-zero variance or strong collinearity. This yields a more stable and compact representation while preserving key structural differences between families.

C. Graphlet Correlation Matrix (GCM)

To obtain graph-level descriptors, we treat the orbits as features across nodes and compute a correlation matrix expressing relationships between orbit counts. The upper triangular portion of this matrix is flattened into a single feature vector. These GCM vectors provide expressive, higher-order structural signatures that capture how different graphlet roles relate within a graph.

IV. Modeling Approaches

A. Coarse Baseline

The baseline uses lightweight structural features: node and edge counts, density, clustering coefficient, summary degree statistics, and basic motifs such as triangles, 3-paths, and small stars. This approach provides an interpretable benchmark for assessing the benefit of graphlet-based methods.

B. Graphlet Benchmark

The benchmark uses the flattened GCM vectors. These features capture orbit interactions and structural dependencies that simple statistics miss, making them a richer representation of call graph topology. This method tests whether graphlet correlations encode family-level structural "signatures."

V. Classification Setup

We evaluate multiple classifiers, including Random Forests, Gradient Boosted Trees, and Support Vector Machines. Training uses either a 60/20/20 split or stratified 5-fold cross-validation, ensuring class proportions are preserved. All splits occur at the graph level to avoid leakage between training and test samples. We also explore whether models trained on coarse features behave differently from those trained on GCM vectors, and whether the added structural detail improves generalization.

VI. Evaluation

We compute accuracy, macro-F1, macro-AUC, and confusion matrices on held-out data. ROC curves using prediction scores allow us to assess ranking performance. Comparisons between the coarse and GCM pipelines show how much structural information each feature set captures. We also examine consistency across folds, analyze common misclassifications, and inspect feature importances from tree-based models to identify which orbit correlations carry the strongest discriminative signal.

VII. Team Contributions

Hernan Barajas: GDV/GCM pipeline, baseline engineering, classifier implementation, evaluation. **Tom O'Connor:** Visualization, experimental design, ROC/metric analysis, interpretation, and writing.